

Adroit Technologies

Protocol Driver

Name	Mitsubishi FX3 Ethernet TCP/IP
DLL	MITFXETH



COPYRIGHT

Copyright © 2022 Advanced Worx 112 (Pty) Ltd. All rights reserved.

Information in this document is subject to change without notice. The software described in this document is furnished under a license agreement or nondisclosure agreement. The software may be used or copied only in accordance with the terms of those agreements. No part of this publication may be reproduced, stored in a retrieval system, or transmitted in any form or any means electronic or mechanical, including photocopying and recording for any purpose other than the purchaser's personal use without the written permission of Advanced Worx 112 (Pty) Ltd.

Advanced Worx 112 (Pty) Ltd
20 Waterford Office Park
189 Witkoppen Road (c/o Waterford Drive)
Fourways
South Africa

<https://adroittech.co.za/>

Contents

1. Overview	4
1.1. PLC software.....	4
1.1.1. Fx3U Series PLC setup.....	4
2. Wiring	5
3. Device Configuration	6
3.1. Device Configuration	6
3.1.1. Port Selection	6
3.1.2. PLC Details.....	6
3.1.3. Local Details.....	7
3.1.4. mServer Details	7
3.1.5. Test Device Configuration Connectivity	7
4. Supported Addresses	9
4.1. Mitsubishi FX3U Series PLC	9
5. Protocol Definition and Considerations	10
5.1. Data Registers	10
5.2. Link Registers	10
5.3. File Registers	10
5.4. Extended File Registers	11
5.5. Timers.....	11
5.6. Counters.....	11
5.7. Inputs	12
5.8. Outputs	12
5.9. Internal Relay	12
5.10. Link Relay.....	14
5.11. Annunciators.....	14
6. General Notes and Troubleshooting	15
6.1. Contact Details.....	15
6.2. Adroit Technologies Knowledge Base.....	15
6.3. Troubleshooting	15
7. Appendix A.....	16
7.1. How to configure the FX3U Ethernet module for communication to Adroit	16
7.1.1. Step 1:	16
7.1.2. Step 2:	17
7.1.3. Step 3:	18
7.1.4. Step 4:	19
7.1.5. Step 5:	19
7.1.6. Step 6:	20
8. Revisions.....	20
8.1. Document Revision.....	20
8.2. DLL Revision	20

1. Overview

This driver supports the FX3U PLC on Ethernet.

1.1. PLC software

1.1.1. Fx3U Series PLC setup

FX Configurator-EN software is required to allow communication with the FX3U Ethernet card.

[Appendix A](#) has step by step instructions on how to setup the Ethernet card that Adroit can talk to it using the FX Configurator-EN software.

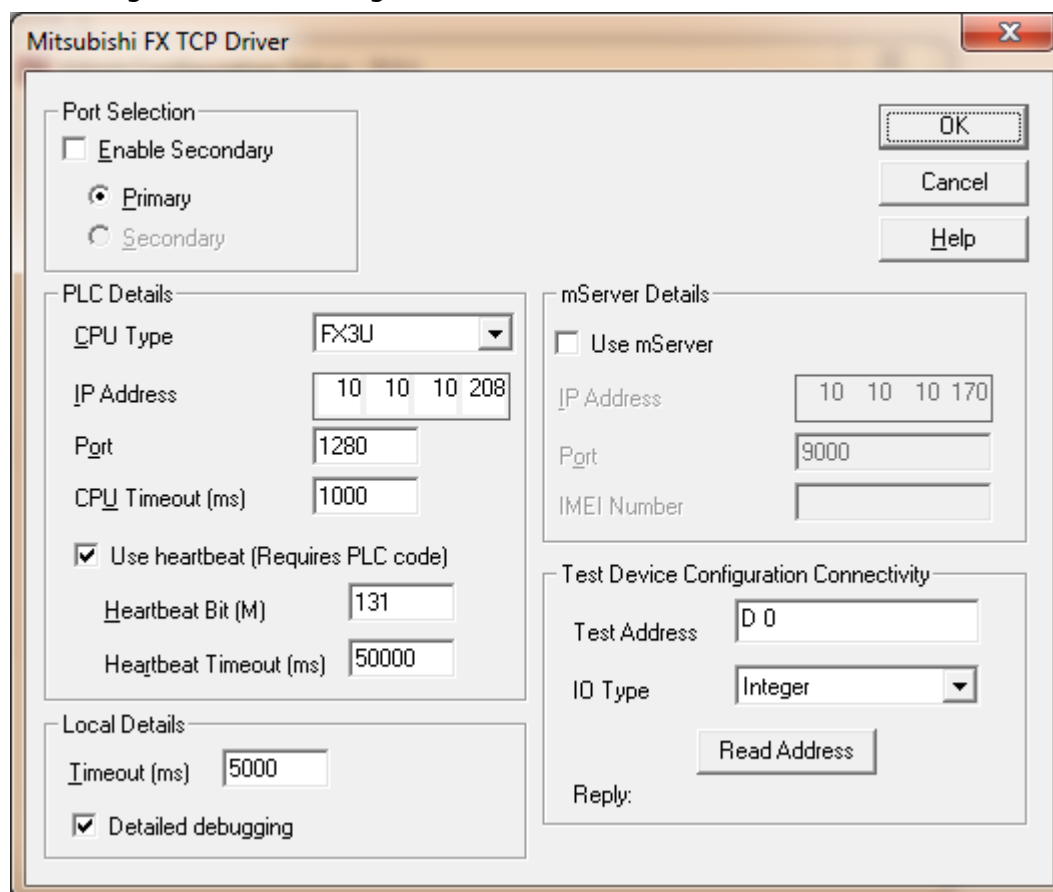
2. Wiring

Standard Ethernet.

3. Device Configuration

3.1. Device Configuration

One Device Agent must be configured for each CPU to be communicated to.



3.1.1. Port Selection

Enable Secondary – Enables secondary configuration for dual redundant PLCs. This setting is disabled for this driver.

Primary / Secondary – Select the primary or secondary button depending on which connection you wish to configure.

3.1.2. PLC Details

CPU Type - The following CPU types are supported: FX3U.

IP Address - Identifies the Mitsubishi PLC's IP addresses. This address must be unique. It defaults to your local IP (40.0.0.1).

Port - TCP/IP port number. The PLC software shipped with the driver is programmed for port 1280 - 1287.

CPU Time-out - This is the time-out period for transactions between the AJ71E71 card and the PLC CPU.

Use heartbeat (Requires PLC code) - Enables the use of the heartbeat functionality to check for connection availability.

Heartbeat Bit (M) - In order to guarantee the re-establishment (under all circumstances) of a connection between Adroit and the Mitsubishi PLC, the heartbeat bit is used. Adroit will set this bit every Heartbeat time-out period. The PLC program will reset this bit immediately after Adroit has set it. The PLC program will also monitor this bit and reset the connection if the 120 seconds (PLC program default) has elapsed without being set by Adroit.

Heartbeat Timeout - The heartbeat pulse generated by Adroit to keep the communication link alive. Adroit will set the Heartbeat Bit every Heartbeat time period.

3.1.3. Local Details

Timeout (ms) - The maximum time allowed for Adroit to PLC communication messages before a time-out error will be registered.

Detailed debugging - Currently detailed debugging does the following:

1. Shows transaction start and end times
2. Describes each read operation before executing it

The Agent Server must as always be restarted for the options changed in this dialog to take effect.

3.1.4. mServer Details

This section is only applicable if the mServer software is used in conjunction with eTung GPRS routers.

Use mServer – Directs the driver to use the mServer to look up the IP address of the PLC from the mServer.

IP Address – The IP Address of the mServer.

Port – The port that the mServer uses to communicate.

IMEI Number – The IMEI Number of the router that the mServer needs to provide the IP Address for.

3.1.5. Test Device Configuration Connectivity

This section is available to help users determine whether or not the settings they have entered are valid and Adroit can communicate with your device.

Test Address – An address in your device which will be polled when the "Read Address" button is clicked. The address entered must be valid.

IO Type – The address type being read. i.e. Boolean, integer, real or string.

Read Address – Click this button to poll your device with the current settings in your dialog.

Reply – When "Read Address" is clicked, Adroit will poll your device. The value returned from your device will be displayed here.

4. Supported Addresses

4.1. Mitsubishi FX3U Series PLC

Mitsubishi FX3U series PLC data (device) types supported in the Adroit Protocol.



PLEASE NOTE

1. Only CN and TN registers are supported by using the C and T address type codes respectively. CS and TS registers are not supported at this time.

2. There is not support for the M8000 to M8511 range of registers at this time.

Please see Table on page 9-17 of the Ethernet Interface Module Manual

Device		Min Range	Max Range*	PLC Data Type	Boolean	Integer	Real	String
Data Reg	D	0	7999	Int 16	D 1.2	D 0	D 0	D 0 Sn
			7999	BCD 16		D 0 B	D 0 B	
			7998	BCD 32		D 0 C	D 0 C	
			7998	Long		D 0 L	D 0 L	
			7998	Real 32			D 0 R	
		8000	8511	Int 16	D 1.2	D 0	D 0	D 0 Sn
			8511	BCD 16		D 0 B	D 0 B	
			8510	BCD 32		D 0 C	D 0 C	
			8510	Long		D 0 L	D 0 L	
			8510	Real 32			D 0 R	
File Reg	R	0	32767	Int 16	R 0.0	R 0	R 0	R 0 Sn
			32767	BCD 16		R 0 B	R 0 B	
			32766	BCD 32		R 0 C	R 0 C	
			32766	Long		R 0 L	R 0 L	
			32766	Real 32			R 0 R	
Timer (TN only)	T	0	511	Int 16		T 0	T 0	
Counter (CN only)	C	0	255	Int 16		C 0	C 0	
Inputs	X	0	377	Bool	X 0	X 0 (16)	X 0 (16)	
			377	BCD 16		X 0 B	X 0 B	
Outputs	Y	0	377	Bool	Y 0			
State	S	0	4095	Bool	S 0			
Internal Relay	M	0	7679	Bool	M 0			
Not supported	M	8000	8511	Bool	M 0			

Address register numbers are in decimal or HEX notation depending on:

Device	Prefix	Notation
Data Reg	D	DECIMAL
Link Reg	W	HEX
File Reg	R	DECIMAL
Ext File Reg	E	DECIMAL
Timer	T	DECIMAL
Counter	C	DECIMAL
Inputs	X	HEX
Outputs	Y	HEX
Internal Relay	M,L,S	DECIMAL
Link Relay	B	HEX
Annunciator	F	DECIMAL

5. Protocol Definition and Considerations

5.1. Data Registers

Read request:

Read word	PC CPU	2500 ms t/o	Offset	Type	Length	End
01	ff	0a 00	00 00 00 00	20 44	00	00

Read response:

Subheader	Completion Code	First data word	Last data word
81	00	00 00	00 00

Write request:

Write word	PC CPU	2500 ms t/o	Offset	Type	Length	End
03	ff	0a 00	00 00 00 00	20 44	00	00

First word	Last Word
00 00	00 00

Write response:

Subheader	Completion Code
83	00

5.2. Link Registers

Read request:

Read word	PC CPU	2500 ms t/o	Offset	Type	Length	End
01	ff	0a 00	00 00 00 00	20 57	00	00

Read response:

Subheader	Completion Code	First data word	Last data word
81	00	00 00	00 00

Write request:

Write word	PC CPU	2500 ms t/o	Offset	Type	Length	End
03	Ff	0a 00	00 00 00 00	20 57	00	00

First word	Last Word
00 00	00 00

Write response:

Subheader	Completion Code
83	00

5.3. File Registers

Read request:

Read word	PC CPU	2500 ms t/o	Offset	Type	Length	End
01	ff	0a 00	00 00 00 00	20 52	00	00

Read response:

Subheader	Completion Code	First data word	Last data word
81	00	00 00	00 00

Write request:

Write word	PC CPU	2500 ms t/o	Offset	Type	Length	End
03	ff	0a 00	00 00 00 00	20 52	00	00

Write response:

Subheader	Completion Code
83	00

5.4. Extended File Registers

Read request:

Read word	PC CPU	2500 ms t/o	Offset	Type	Block	Length	End
17	ff	0a 00	00 00 00 00	20 52	01 00	00	00

Read response:

Subheader	Completion Code	First data word	Last data word
97	00	00 00	00 00

Write request:

Write word	PC CPU	2500 ms t/o	Offset	Type	Block	Length
18	ff	0a 00	00 00 00 00	20 52	01 00	00

End	First word	Last Word
00	00 00	00 00

Write response:

Subheader	Completion Code
98	00

5.5. Timers

Read request (Present Value):

Read word	PC CPU	2500 ms t/o	Offset	Type	Length	End
01	ff	0a 00	00 00 00 00	4e 54	00	00

Response:

Subheader	Completion Code	First data word	Last data word
81	00	00 00	00 00

Write request:

Write word	PC CPU	2500 ms t/o	Offset	Type	Length	End
03	ff	0a 00	00 00 00 00	4e 54	00	00

First word	Last Word
00 00	00 00

Write response:

Subheader	Completion Code
83	00

5.6. Counters

Read request (Present Value):

Read word	PC CPU	2500 ms t/o	Offset	Type	Length	End
01	ff	0a 00	00 00 00 00	4e 43	00	00

Response:

Subheader	Completion Code	First data word	Last data word
81	00	00 00	00 00

Write request:

Write word	PC CPU	2500 ms t/o	Offset	Type	Length	End
03	ff	0a 00	00 00 00 00	4e 43	00	00

First word	Last Word
00 00	00 00

Write response:

Subheader	Completion Code
83	00

5.7. Inputs

Read request:

Read bit	PC CPU	2500 ms t/o	Offset	Type	Length	End
00	ff	0a 00	00 00 00 00	20 58	00	00

Read response:

Subheader	Completion Code	First data byte	Last data byte
80	00	00	00

Write request:

Write word	PC CPU	2500 ms t/o	Offset	Type	Length	End
02	ff	0a 00	00 00 00 00	20 58	00	00

First byte	Last byte
00	00

Write response:

Subheader	Completion Code
82	00

5.8. Outputs

Read request:

Read bit	PC CPU	2500 ms t/o	Offset	Type	Length	End
00	ff	0a 00	00 00 00 00	20 59	00	00

Read response:

Subheader	Completion Code	First data byte	Last data byte
80	00	00	00

Write request:

Write bit	PC CPU	2500 ms t/o	Offset	Type	Length	End
02	ff	0a 00	00 00 00 00	20 59	00	00

First byte	Last byte
00	00

Write response:

Subheader	Completion Code
82	00

5.9. Internal Relay

Read request:

Read bit	PC CPU	2500 ms t/o	Offset	Type	Length	End
00	ff	0a 00	00 00 00 00	20 4d	00	00

Read response:

Subheader	Completion Code	First data byte	Last data byte
80	00	00	00

Write request:

Write bit	PC CPU	2500 ms t/o	Offset	Type	Length	End
02	ff	0a 00	00 00 00 00	20 4d	00	00

First byte	Last byte
00	00

Write response:

Subheader	Completion Code
82	00

5.10. Link Relay

Read request:

Read word	PC CPU	2500 ms t/o	Offset	Type	Length	End
01	ff	0a 00	00 00 00 00	20 42	01	00

Read response:

Subheader	Completion Code	First data byte	Last data byte
81	00	00	00

Write request:

Write bit	PC CPU	2500 ms t/o	Offset	Type	Length
02	ff	0a 00	00 00 00 00	20 42	01

First byte	Last byte	End
00	00	00

Write response:

Subheader	Completion Code
82	00

5.11. Annunciators

Read request:

Read word	PC CPU	2500 ms t/o	Offset	Type	Length	End
01	ff	0a 00	00 00 00 00	20 46	01	00

Read response:

Subheader	Completion Code	First data byte	Last data byte
81	00	00	00

Write request:

Write bit	PC CPU	2500 ms t/o	Offset	Type	Length
02	ff	0a 00	00 00 00 00	20 46	01

First byte	Last byte	End
00	00	00

Write response:

Subheader	Completion Code
82	00

6. General Notes and Troubleshooting

6.1. Contact Details

Email: support@adroit.co.za and drivers@adroit.co.za

Phone: +27 11 658-8100

Web: www.adroit.co.za

6.2. Adroit Technologies Knowledge Base

For additional information, please refer to our Knowledgebase website:

[Latest Drivers topics - Features, discussions, tips, tricks, questions, problems and feedback \(adroittechnologiesautomation.com\)](http://adroittechnologiesautomation.com)

6.3. Troubleshooting

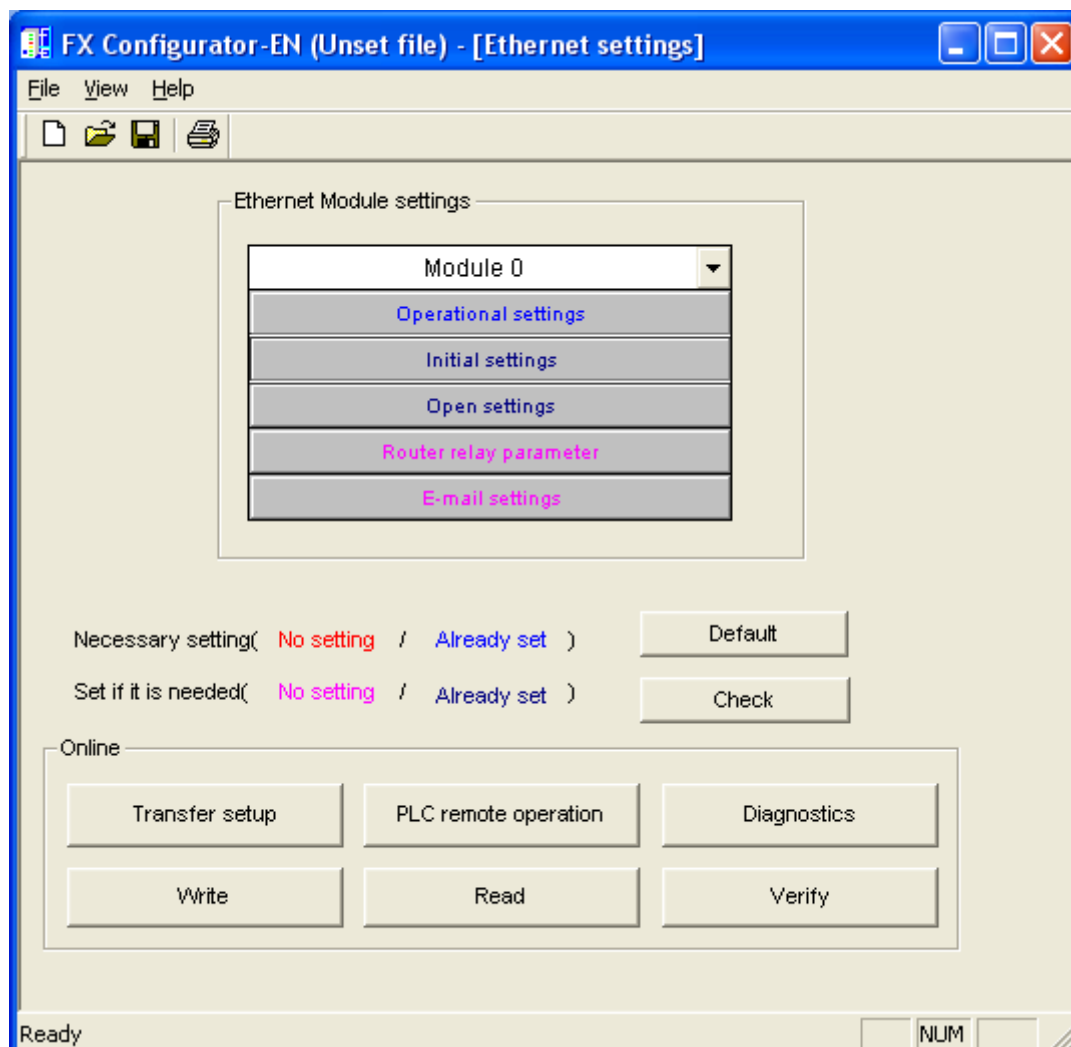
It is possible that by modifying the default settings, that the **watchdog** sent by the driver may not arrive in the PLC, thus causing the PLC to “intermittently” reset our connection to the PLC. Ensure that the timer in the PLC (which runs for 120 seconds by default) is at least double the setting used by the driver. If the driver channel is backlogged (i.e. the driver is too busy with polling tasks), the watchdog may also be delayed, as it is not a foreground process in the driver. Driver backlogging or queuing is easily detectable by setting a value in register in the PLC, and then timing how long it takes to receive a feedback via another tag which was scanned to the same register.

7. Appendix A

7.1. How to configure the FX3U Ethernet module for communication to Adroit

7.1.1. Step 1:

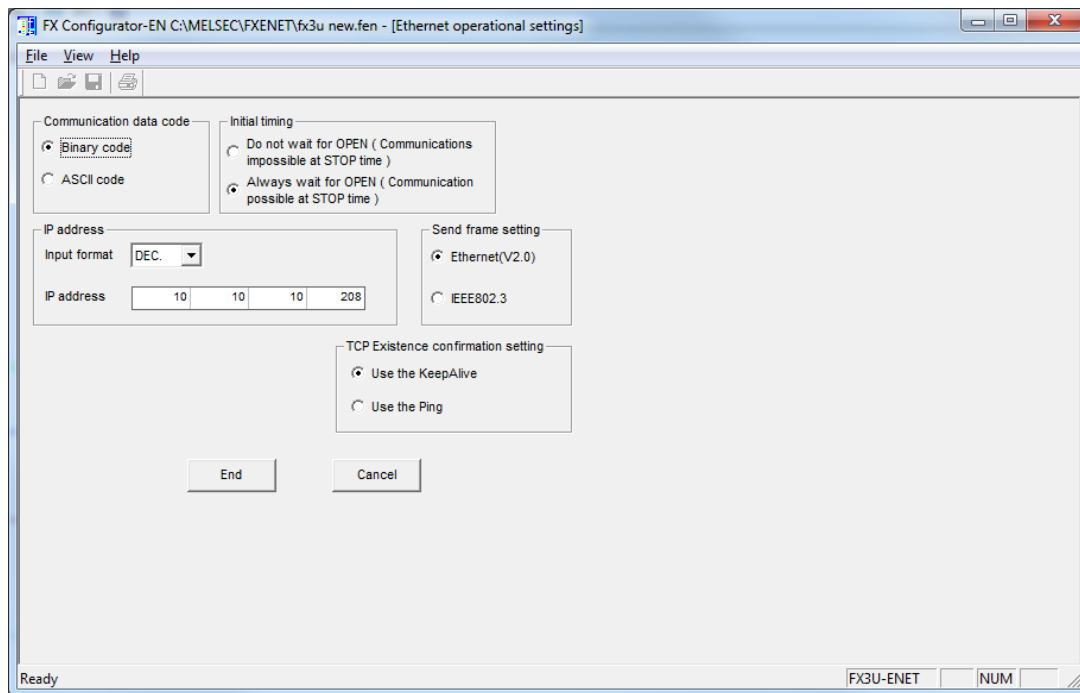
When starting up the FX Configurator-EN software you will be represented with the following screen. Under Ethernet Module Settings there is a drop down box to select which module your Ethernet card is connected as. "Module 0" selection means it is the first card after the CPU.



7.1.2. Step 2:

Click on the "Operational Settings" button to get the next screen.

In the operational settings screen, the settings should look exactly as below, except for your IP address that might differ. This will be your IP address you want to configure the Ethernet card for. Once completed press "END" to take you back to the main screen.

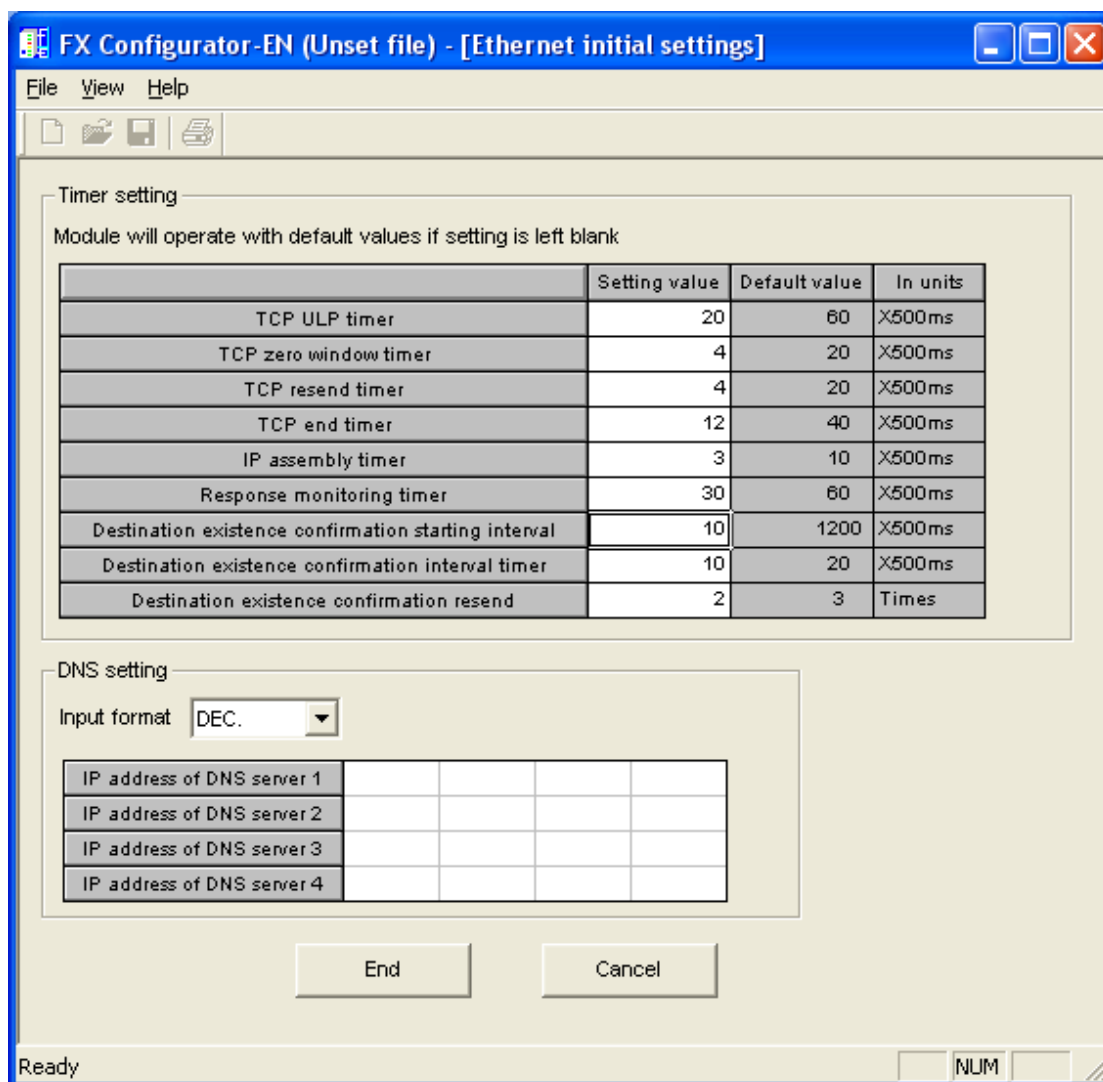


7.1.3. Step 3:

Click on the "Initial Settings" button to get to the next screen.

In the initial settings screen, the settings should look exactly as below.

Once completed press "END" to take you back to the main screen.



FX Configurator-EN (Unset file) - [Ethernet initial settings]

File View Help

Timer setting

Module will operate with default values if setting is left blank

	Setting value	Default value	In units
TCP ULP timer	20	60	X500ms
TCP zero window timer	4	20	X500ms
TCP resend timer	4	20	X500ms
TCP end timer	12	40	X500ms
IP assembly timer	3	10	X500ms
Response monitoring timer	30	60	X500ms
Destination existence confirmation starting interval	10	1200	X500ms
Destination existence confirmation interval timer	10	20	X500ms
Destination existence confirmation resend	2	3	Times

DNS setting

Input format: DEC.

IP address of DNS server				
IP address of DNS server 1				
IP address of DNS server 2				
IP address of DNS server 3				
IP address of DNS server 4				

End Cancel

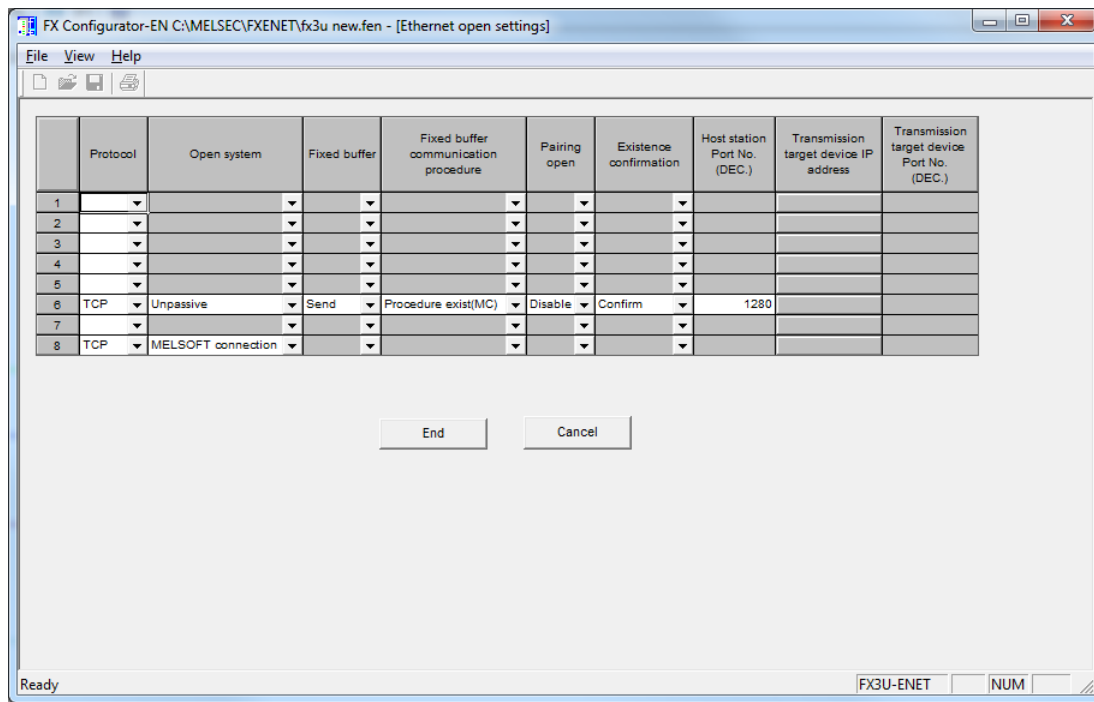
Ready NUM

7.1.4. Step 4:

Click on the "Open Settings" button to get to the next screen.

In the open settings screen, the settings should look exactly as below, except for your Port which you can configure to be what you require the card port to be.

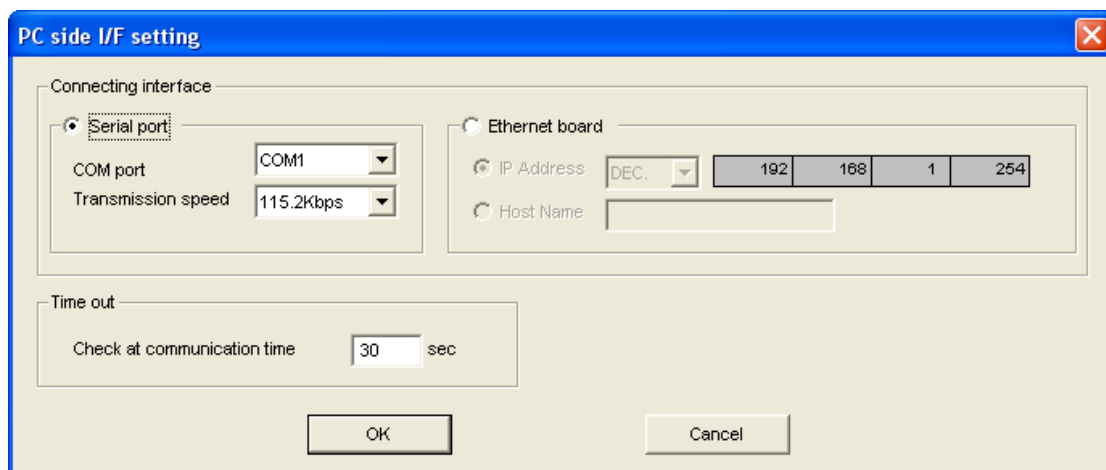
Once completed press "END" to take you back to the main screen.



7.1.5. Step 5:

Click on the "Transfer Setup" button to get to the next screen.

In this screen, you select which method to use download the setup to the PLC. As we only configured the Ethernet module now, we will select "Serial Port" and press ok to take you back to the main screen. You will require the "RED" Mitsubishi Beijer SC-09 serial cable to do the download.

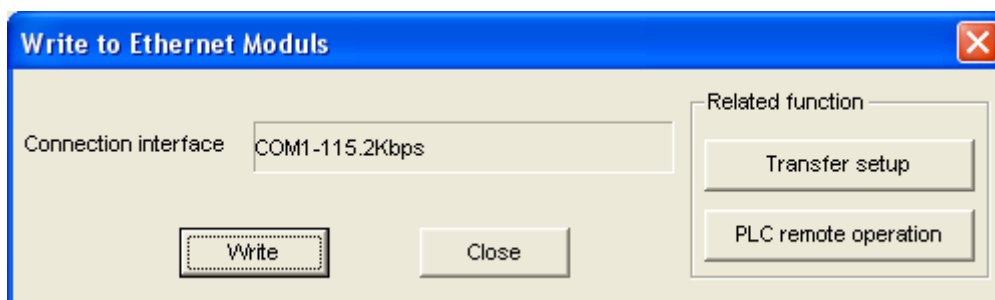


7.1.6. Step 6:

Click on the "Write" button to get to the next screen.

In this screen, you just click "Write" and it will start to download the setup to the PLC.

It should return with a message box that says "completed". Press OK and close the whole software package. Your Fx3U Ethernet card will not be set.



8. Revisions

8.1. Document Revision

Rev.	Date	Author	Comment	Approval
1.0	14-Dec-2022	J. Duncan	New Format	D. Jordaan
1.1	25-Jan-2023	K. Robinson	Changes to format, updated image resources.	D. Jordaan
2.0				

8.2. DLL Revision

Rev.	Date	Changes
1	08/02/2012	Morphed from AJE71 Driver and removed all A Series from Driver as this is now a FX PLC only driver
110	10/05/2012	Added support for eTung GPRS routers and mServer software.
117	14/06/2013	Strings were not sized correctly, if anything smaller than S80 , fixed. Reads and writes needed to have a purge if residual bytes were detected on the socket Odd string lengths have been disabled.